Mutual Fund Analytics Capstone Project

Project Title:

Mutual Fund Analytics Platform
Advanced Data Analytics & Business Intelligence Capstone Project

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Abstract

The Mutual Fund Analytics Platform is a comprehensive financial analytics project designed to transform raw mutual fund data into meaningful investment insights. The project integrates Python, SQLite, SQL, and Power BI to build an end-to-end analytics pipeline covering data ingestion, cleaning, exploratory analysis, financial performance evaluation, portfolio analytics, benchmark comparison, and advanced risk analysis.

The project analyzes historical Net Asset Value (NAV), investor transactions, fund performance, benchmark indices, portfolio holdings, and fund house information. Advanced financial metrics including CAGR, Sharpe Ratio, Sortino Ratio, Alpha, Beta, Maximum Drawdown, and Value at Risk (VaR) were computed to evaluate investment performance and risk. Interactive Power BI dashboards were developed to visualize industry trends, investor behavior, portfolio allocation, and benchmark performance.

The project demonstrates how data analytics and business intelligence can assist investors in making informed mutual fund investment decisions.

Introduction

Mutual funds have become one of the most preferred investment options due to their diversification and professional management. However, investors often struggle to compare schemes, evaluate historical performance, measure risk, and identify suitable investment opportunities.

This project addresses these challenges by building a complete Mutual Fund Analytics Platform capable of analyzing financial datasets and presenting actionable insights through interactive dashboards and advanced analytics techniques.

Problem Statement

The mutual fund industry generates a significant volume of financial data every day. Investors require efficient tools to analyze fund performance, compare schemes, evaluate risk, and monitor investment portfolios. Traditional spreadsheet-based analysis becomes difficult as data volume increases.

Therefore, an integrated analytics platform is required to automate data processing, calculate financial metrics, and provide meaningful visualizations for investment decision-making.

Objectives

- Analyze historical NAV data
- Study investor behavior
- Evaluate mutual fund performance
- Calculate financial risk metrics
- Develop interactive Power BI dashboards
- Build a recommendation engine
- Perform Value at Risk (VaR) analysis
- Generate actionable investment insights

Dataset Description

The project uses ten datasets containing mutual fund information.

Dataset	Description
Fund Master	Scheme information
NAV History	Historical NAV values
AUM by Fund House	Assets under Management
Monthly SIP Inflows	SIP investments
Category Inflows	Category-wise investments
Industry Folio Count	Investor folio statistics
Scheme Performance	Returns and ratios
Investor Transactions	Transaction history
Portfolio Holdings	Equity holdings
Benchmark Indices	Market benchmark data

Dataset Statistics

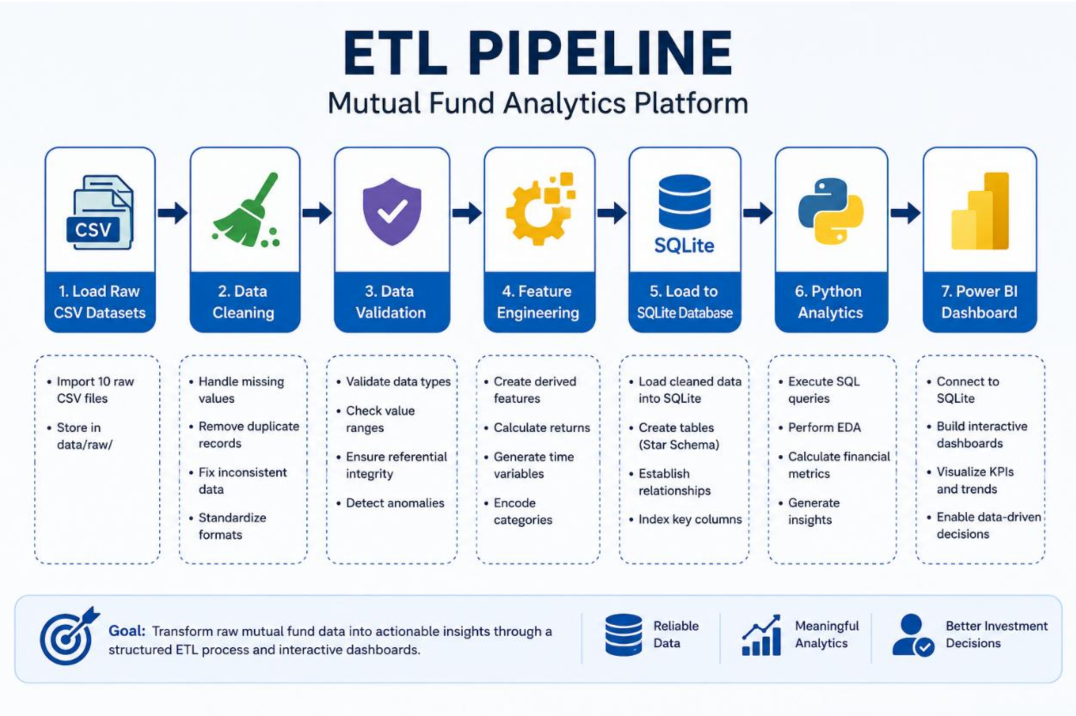
- NAV Records: **46,000**
- Investor Transactions: **32,778**
- Mutual Fund Schemes: **40**
- Portfolio Holdings: **322**
- Benchmark Records: **8,050**

- Unique Investors: **5,000**

Technology Stack

Technology	**Purpose**
Python	Data Analysis
Pandas	Data Processing
NumPy	Numerical Computation
Matplotlib	Visualization
SQLite	Database
SQL	Data Querying
Power BI	Dashboard Development
Git & GitHub	Version Control
Jupyter Notebook	Analysis Environment

ETL Pipeline



The ETL process consists of the following stages:

1. Load raw CSV datasets

2. Clean missing and duplicate records
3. Validate data quality
4. Transform variables
5. Load cleaned data into SQLite
6. Perform analytical queries
7. Generate dashboards

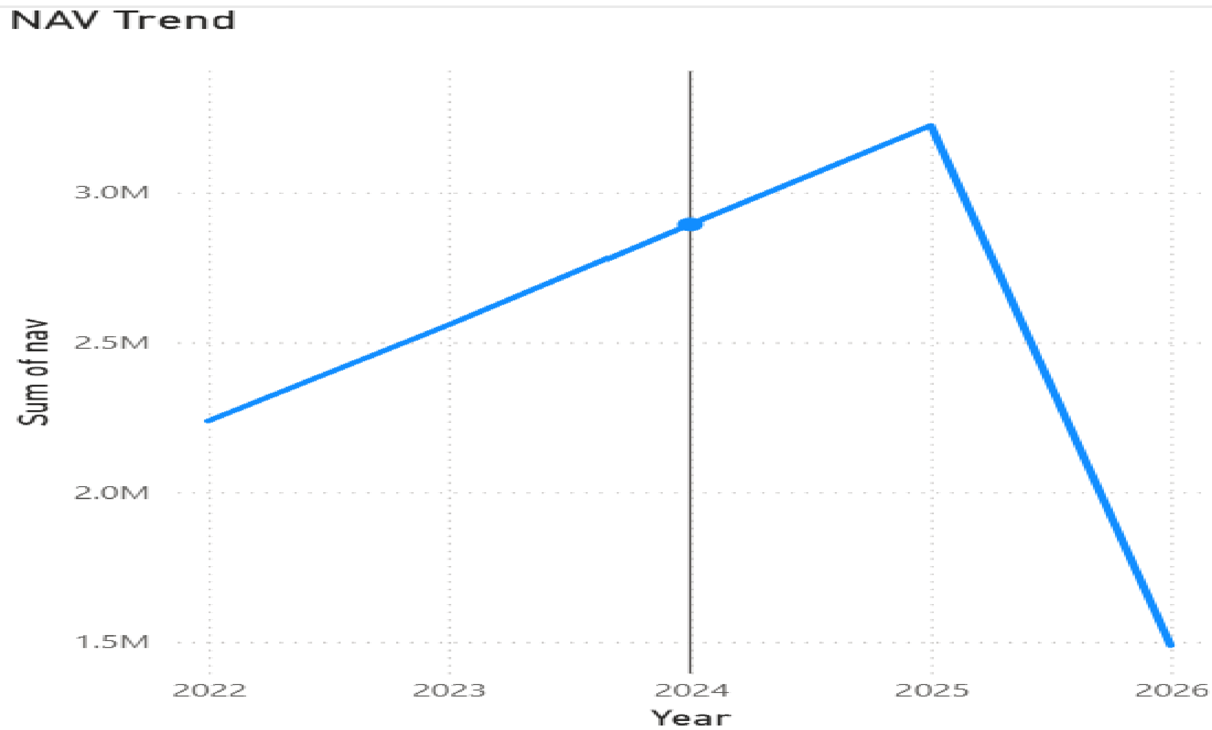
Exploratory Data Analysis

EDA was performed to understand trends across funds, categories, and investors.

Key analyses include:

- NAV Trend
- Category-wise Returns
- Fund House Comparison
- Investor Demographics
- Portfolio Allocation

NAV Trend



Performance Analytics

Several financial metrics were computed to evaluate mutual fund performance.

These include:

- CAGR
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Maximum Drawdown
- Standard Deviation

These metrics provide insights into both return potential and investment risk.

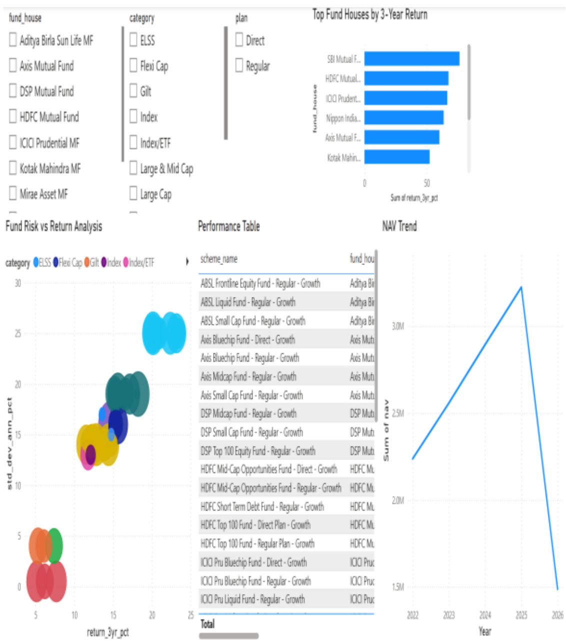
Power BI Dashboard

An interactive Power BI dashboard consisting of five pages was developed.

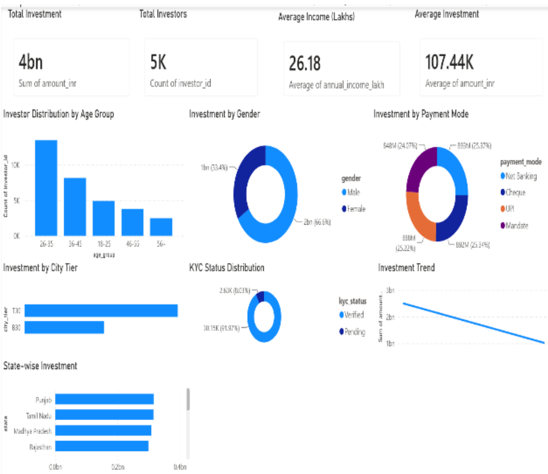
The dashboard includes:

Industry Overview

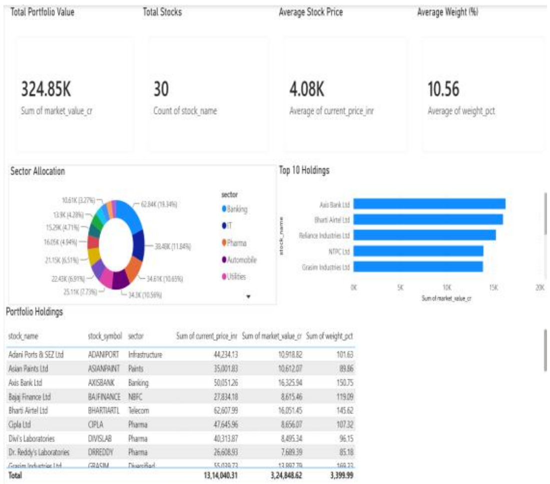
Fund Performance



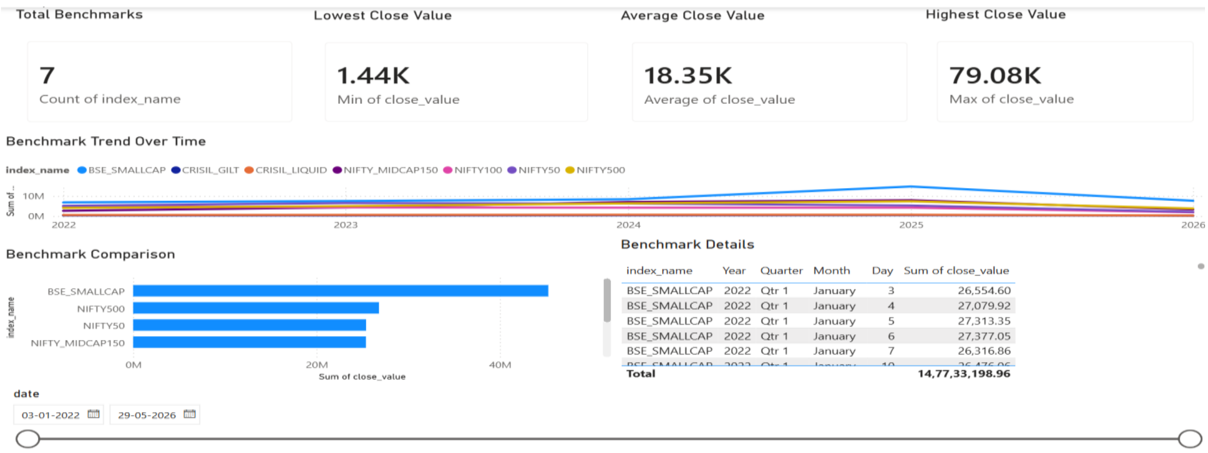
Investor Analytics



Portfolio Holdings



Benchmark Analysis



Advanced Analytics

Advanced analytical techniques implemented include:

Value at Risk (VaR)

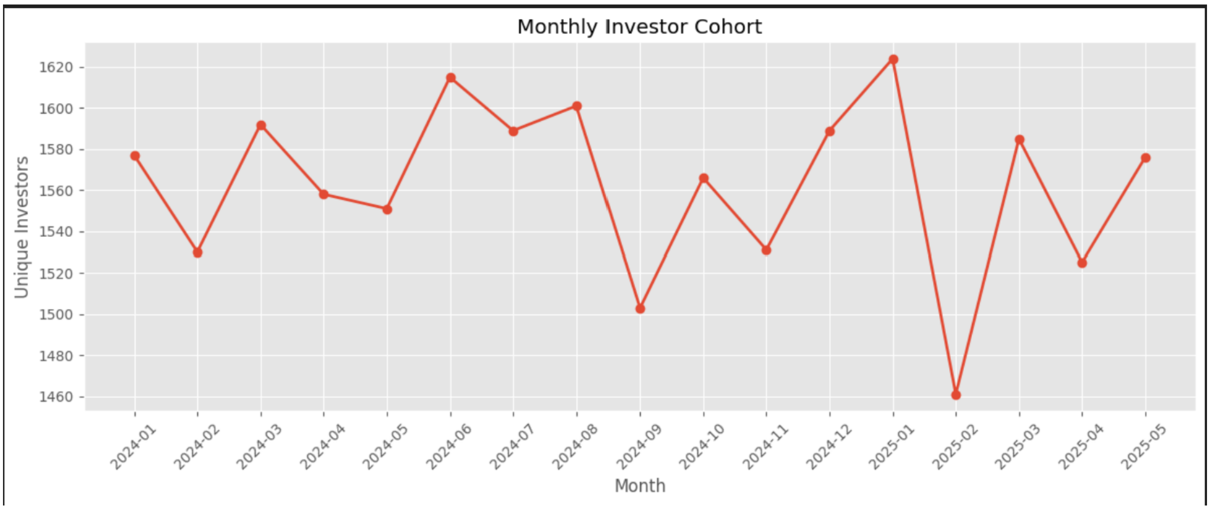
Historical daily returns were used to estimate the 95% Value at Risk.

Result:

95% VaR = -1.63%

Monthly Cohort Analysis

Investor activity was grouped by month to analyze participation trends and retention.



IMG: Monthly Investor Cohort

Recommendation Engine

A recommendation score was calculated using:

- 3-Year Return
- Sharpe Ratio
- Annual Volatility

The system ranks mutual funds based on risk-adjusted performance.

Key Findings

- Large Cap funds demonstrated stable long-term performance.
- Liquid funds exhibited the lowest investment risk.
- Investors aged 26–35 contributed the highest investment volume.
- Verified KYC investors accounted for the majority of transactions.
- Recommendation scores successfully identified top-performing funds.
- Interactive dashboards simplified financial analysis and decision-making.

Conclusion

This project successfully demonstrates how financial analytics, Python programming, SQL databases, and Power BI dashboards can be integrated into a complete business intelligence solution.

The platform transforms raw mutual fund data into actionable insights that support better investment decisions through quantitative analysis, visualization, and risk assessment.

Future Scope

Potential enhancements include:

- Live Mutual Fund API integration
- AI-based recommendation engine
- Streamlit web application
- Portfolio optimization using Modern Portfolio Theory
- Monte Carlo simulation for NAV forecasting
- Automated ETL scheduling
- Weekly email performance reports

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